

MS Algebra Exercises

Neil Makur

Convince Yourself That...

These problems are fairly straightforward and should help you internalize definitions. Convince yourself of each of the following facts:

- A homomorphism $\phi : G \rightarrow H$ is injective if and only if $\ker \phi = 1$
- The action $G \curvearrowright G$ via $g \cdot h = ghg^{-1}$ is actually an action
- If $\varphi_g(h) = ghg^{-1}$ then $\varphi : G \rightarrow \text{Aut}(G)$ is a homomorphism and $\text{im } \varphi \leq \text{Aut}(G)$
- If $\phi : G \rightarrow H$ is a bijective homomorphism, then $\phi^{-1} : H \rightarrow G$ is a homomorphism
- For any group G and fixed element $g \in G$, there is a unique homomorphism $\mathbb{Z} \rightarrow G$ given by $n \mapsto g^n$
- $Z(G)$ is the set of fixed points of the action $G \curvearrowright G$ by conjugation
- $H \leq N_G(H)$
- There are groups $K \leq H \leq G$ with $K \trianglelefteq H$ and $H \trianglelefteq G$ but $K \not\trianglelefteq G$
- $Z_G(g) \leq G$, $N_G(H) \leq G$, and $Z(G) \trianglelefteq G$. Prove this both using their definitions as stabilizers/kernels, and directly using the set-builder definitions

Prime-Power Subgroups

Problem. Let G be a finite group. Prove the *class equation*:

$$|G| = |Z(G)| + \sum_{i=1}^r [G : Z_G(g_i)].$$

Here, there are r non-trivial conjugacy classes (a conjugacy class is a set $\{hgh^{-1} : h \in H\}$, and a trivial conjugacy class has size 1), and g_i is a representative of the i th nontrivial conjugacy class.

Problem. Let G be a finite group with $|G|$ a power of p (where p is a prime), and let $G \curvearrowright X$. Show that $|X| \equiv |\text{Fix}(G)| \pmod{p}$.

Problem. Let G be a finite group with $|G|$ a power of p (where p is a prime). Show that $Z(G) \neq 1$.

Problem. Let p be a prime. Show that all groups of order p^2 are abelian.

Problem. Let G be a finite group and $H \leq G$ a subgroup with $|H|$ a power of the prime p . Show that $[G : H] \equiv [N_G(H) : H] \pmod{p}$.

Problem. For this problem, take the following fact for granted: if G is a finite group and $N \trianglelefteq G$, then there is an order-preserving bijection between subgroups of G/N and subgroups of G that contain N . Further, if $H \leq G/N$ corresponds to $N \leq K \leq G$, then $|H| = [K : N]$. We'll fix a prime p .

- (a) If $H \leq G$ with $p \mid [G : H]$, then there is $H \leq K \leq G$ with $|K| = p \cdot |H|$. Note that H need not be normal in G !
- (b) If $|G| = p^a \cdot m$ for $p \nmid m$, then there is $H \leq G$ with $|H| = p^a$.

(c) If $|G| = p^a$, then we can find subgroups

$$1 \leq G_1 \leq \cdots \leq G_a = G$$

with $|G_i| = p^i$.

Problem. A group is said to be *simple* if it contains no nontrivial normal subgroups. That is, G is simple if the only normal subgroups of G are 1 and G . Fix (not necessarily distinct) primes p and q . Show that no group of order pq is simple.

Counting With Group Actions

Problem (Burnside's Lemma). Let G be a group acting on a set X . Show that the number of orbits of this action is equal to $\frac{1}{|G|} \sum_{g \in G} |\text{Fix}(g)|$ (where $\text{Fix}(g) = \{x : gx = x\}$). Hint:

Problem. Suppose we have n possible colors for each of the vertices of a square. We consider two colorings of the square to be the same if one can be rotated to get the other. How many colorings of the square are there? What if we consider two colorings to be the same if one can be rotated or reflected to get the other? If you feel like it, replace "square" with k -gon.

Problem. How many ways are there to color the squares of an 8×8 chessboard either black or white? Two colorings are the same if one can be rotated to get another. What if we require an equal number of black and white squares?

The Symmetric Group

Problem. Prove that any $\sigma \in S_n$ has a well-defined cycle decomposition by considering the orbits of the action $\mathbb{Z} \curvearrowright [n]$ via $k \cdot n = \sigma^k(n)$.

Problem. Prove that S_n is generated by the two-cycles (i.e. by the permutations that swap two elements). Prove that S_n is actually generated by only $(12), (23), \dots, (n-1, n)$.

Problem. Prove that S_n is generated by (12) and $(23 \cdots n)$.

Problem. Let $\sigma, \rho \in S_n$. Describe the cycle decomposition of $\rho\sigma\rho^{-1}$ in terms of the cycle decomposition of σ .

Problem. For $\sigma \in S_n$, describe the order of σ in terms of the lengths in its cycle decomposition.

Misc

Problem. For any $a_1, \dots, a_n \in \mathbb{Z}$, describe $\langle a_1, \dots, a_n \rangle \leq \mathbb{Z}$.

Problem. Let G be a finite group. Prove that there are no homomorphisms $G \rightarrow \mathbb{Z}$ except for the trivial homomorphism $g \mapsto 1$.

Problem. Prove the following *universal property* of products: the product $G \times H$ has natural maps $\pi_G : G \times H \rightarrow G$ and $\pi_H : G \times H \rightarrow H$. For any other group K with maps $f_G : K \rightarrow G$ and $f_H : K \rightarrow H$, there is a unique $\phi : K \rightarrow G \times H$ with $f_G = \pi_G \circ \phi$ and $f_H = \pi_H \circ \phi$.

Problem. Let G be a group with $\text{Aut}(G) = 1$. What can you say about G ?